Chia-Hsiang Kao

World Models · Video Generation · LLM Agents

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SUMMARY

Cornell CS Ph.D. (expected Fall 2027) building world models that recover and predict real-world 3D motion and physics, and video generation models that learn new tasks in context. Led all first-author projects, with papers at CVPR, NeurIPS, ICLR, ACL, and MICCAI; 500+ citations (h-index 8). Student Researcher at Meta; previously Applied Scientist Intern at Amazon. Trained as a physician (M.D.).

EXPERIENCE

Student Researcher · Meta, New York, NY — Jan 2026 – May 2027

- Leading **amodal scene flow completion**: jointly completing occluded scene geometry and its 3D motion, beyond visible-only scene flow; designed a novel point-cloud tokenizer and a diffusion model that completes point-voxel tokens while predicting the motion field.
- Led **ViGeo**, a video in-context learning framework that unifies video tasks as spatiotemporal canvas completion; a single model absorbs a diverse task taxonomy spanning camera control, modality-guided generation, and video manipulation, and generalizes to unseen tasks and sensing modalities. Fully fine-tuned Wan2.2-5B on 32 A100 nodes.

Applied Scientist Intern · Amazon, Sunnyvale, CA — May 2025 – Aug 2025

- Led **Δynamics** (CVPR 2026), a VLM that recovers rigid-body physics from monocular video by generating MuJoCo scene configurations as structured text. Trained on 400K synthetic videos, it substantially outperforms frontier VLMs, improves further with test-time search, and transfers to real-world videos.

Graduate Researcher · Cornell University, Ithaca, NY — 2023 – Present

- Led **UnivEarth** (ACL 2026 Findings), a benchmark and harness for code-executing LLM agents on Earth observation; showed frontier agents reach only 33% accuracy as generated code fails 58% of the time. Mentored two junior researchers.
- Led **AllClear** (NeurIPS 2024 D&B), a cloud-removal benchmark for satellite imagery, mentoring one junior researcher; proposed **counter-current learning** (NeurIPS 2024), a biologically plausible alternative to backprop.

SELECTED PUBLICATIONS

Unifying Video Tasks via Spatiotemporal Analogy — **Under review**
Δynamics: Language-Based Representation for Inferring Rigid-Body Dynamics From Videos — **CVPR 2026**
Towards LLM Agents for Earth Observation — **ACL 2026 Findings**
Counter-Current Learning: A Biologically Plausible Dual Network Approach for Deep Learning — **NeurIPS 2024**
AllClear: A Dataset and Benchmark for Cloud Removal in Satellite Imagery (co-first author) — **NeurIPS 2024 D&B**
MAML Is a Noisy Contrastive Learner in the Classification — **ICLR 2022**
Demystifying T1-MRI to FDG18-PET Image Translation via Representational Similarity — **MICCAI 2021**

First author unless noted. Full list of 11 papers on Google Scholar.

EDUCATION

Ph.D., Computer Science · Cornell University — 2023 – Fall 2027 (expected)
Advisor: Bharath Hariharan

Doctor of Medicine (M.D.) · National Yang-Ming Chiao-Tung University — 2015 – 2022

SKILLS, HONORS & SERVICE

Modeling: world models, video generation (flow matching, Wan2.2), VLMs, LLM agents & tool use
Tools: PyTorch, FastVideo, multi-node distributed training, MuJoCo, Google Earth Engine
Honors: Cornell Foundational AI PhD Fellowship (2025), MICCAI Student Travel Award (2021), NSTC Undergraduate Research Fellowship (2018, 2020)
Reviewer: NeurIPS, ICLR, ICML, ECCV, COLM, AAAI, AISTATS; CVIU, IEEE TETCI